

Preprocessing of Medical Images using Deep Learning: A Comprehensive Review

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Abstract— Medical image preprocessing is a vital component in improving the quality and interpretability of medical images, which directly impacts accurate diagnosis and treatment planning. The advancements in deep learning techniques have paved the way for creative solutions to the problems posed by medical image preprocessing. This review article seeks to offer a thorough examination of contemporary deep-learning methods used for the preprocessing of medical images. The study delves into various preprocessing tasks, including denoising, image enhancement, registration, and segmentation, and examines how deep learning techniques have effectively tackled these tasks. It highlights the significant contributions of deep learning-based preprocessing methods in enhancing image quality, reducing noise, improving contrast, and extracting precise anatomical structures. Moreover, the study emphasizes the impact of deep learning-based preprocessing methods on clinical decision-making. By improving the quality and interpretability of medical images, these techniques empower radiologists and clinicians to make accurate diagnoses, plan treatments, and monitor disease progression more effectively. The review also explores the potential applications of deep learning-based preprocessing methods across different clinical imaging practices, including CT, ultrasound, and MRI.

Keywords— *imaging modalities, noise reduction, image segmentation, convolutional neural networks (CNNs), generative adversarial networks (GANs), tumor detection*

I. INTRODUCTION

A. Background

Medical imaging has revolutionized the field of healthcare by providing non-invasive methods for diagnosing and monitoring various diseases and conditions. Clinical decision-making is aided by the useful information contained in clinical pictures like X-rays, CT scans, MRI scans, and ultrasound pictures. However, these images often suffer from noise, artifacts, inconsistencies, and low contrast, which can hinder accurate interpretation and analysis. Preprocessing techniques aim to address these challenges by improving the quality of medical images and extracting relevant information for

subsequent analysis. Diagnostic imaging has traditionally utilised preprocessing techniques like filtering, histogram equalization, and morphological operations. These methods have shown some success, but they often rely on handcrafted features and may not fully exploit the complex patterns and structures present in medical images. A paradigm shift in medical image preprocessing has occurred with the development of deep learning [1]. Convolutional neural networks (CNNs), in particular, have shown exceptional abilities in autonomously acquiring attributes and interpretations straight from the input, improving efficiency in a variety of applications.

B. Motivation

The motivation behind researching the preprocessing of medical images using deep learning stems from the limitations of traditional methods and the advancements made in deep learning techniques. Numerous aspects of computer vision, including picture categorization, object detection, and semantic segmentation, have demonstrated significant promise for deep learning. Applying these techniques to medical image preprocessing can provide several advantages, including improved accuracy, reduced manual intervention, and the ability to handle diverse medical imaging modalities. The motivation also arises from the need to overcome the challenges associated with medical image preprocessing, such as noise removal, enhancing image details, accurately aligning images for registration, and segmenting anatomical structures [2]. Deep learning models have demonstrated the ability to handle these challenges by learning complex representations and patterns directly from the data, making them well-suited for addressing the intricacies of medical images.

C. Objectives

In addition to exploring the developments in deep learning algorithms for denoising, image augmentation, registration, and segmentation tasks, this research study intends to present a thorough review of the problems in medical image preprocessing. The use of deep learning-based preprocessing

for several clinical imaging techniques is investigated, and the effectiveness of these applications is evaluated using the appropriate metrics and benchmark datasets. The study discusses the impact of these techniques on clinical decision-making, identifies limitations and future directions, and aims to equip researchers and practitioners with a comprehensive understanding of deep learning-based preprocessing of clinical images in order to make informed decisions.

II. CHALLENGES IN MEDICAL IMAGE PREPROCESSING

A. Noise Reduction

Noise in medical images can arise from various sources, including sensor imperfections, patient motion, and electronic interference. Noise reduction techniques aim to enhance image quality by reducing unwanted noise while preserving important diagnostic information. Several methods for noise reduction in medical images have been developed in the area of deep learning.

Convolutional neural networks are a popular denoising approach. With the use of CNN-based denoising models, noise is removed from new input photos by learning the underlying noise distribution from pairs of noisy and clean images. These models frequently have an encoder-decoder design with bypass links, allowing the network to collect both local and worldwide picture characteristics. Examples of popular CNN architectures for denoising include U-Net, ResNet, and DenseNet.

Generative adversarial networks (GANs) are a different method for denoising. A discriminator network separates the generated denoised images from the ground truth clean images in GANs, which are made up of a generator network that creates denoised images. By training these networks adversarially, GAN-based denoising models can produce visually pleasing denoised images with enhanced details [3].

B. Image Enhancement

Image-enhancing approaches seek to boost the clarity and interpretability of medical pictures by changing their contrast, brightness, and sharpness. Deep learning algorithms have yielded encouraging results in picture enhancement activities, enabling the generation of enhanced images that highlight important structures and details.

CNNs are used in deep learning-based contrast enhancement techniques to figure out how low-contrast pictures correspond to their enhanced versions. These models learn to amplify the image details while preserving the overall image structure. The networks can be trained on paired low-contrast and high-contrast images or using unsupervised techniques, such as generative models [4]. Super-resolution techniques using deep learning have also been employed for image enhancement. These techniques aim to enhance the resolution and level of detail in low-resolution medical images. High-quality pictures can be produced from low-resolution inputs using CNN architectures, such as SRGAN (Super-Resolution Generative Adversarial Network).

C. Image Registration

Image registration is the act of coordinating and matching numerous medical pictures of identical or distinct individuals in order to compare, assess, and fuse data. By learning spatial transformations and feature representations directly from the data, deep learning-based methods for image registration have demonstrated encouraging results.

Deep learning-based registration methods often involve training a neural network to predict deformation fields or transformation parameters that align the images. These networks are typically trained on pairs of registered images with ground truth transformation information. The network architecture may include convolutional layers for feature extraction and regression layers for predicting the transformation parameters [5].

Evaluation metrics for image registration include voxel-based measures, such as Dice coefficient and mutual information, that quantify the overlap and alignment between registered images. Additionally, visual inspection and clinical validation are crucial for assessing the accuracy and clinical utility of deep learning-based image registration techniques.

D. Image Segmentation

In order to distinguish and identify structures of relevance in medical images, such as tumours, organs, and anatomical regions, image segmentation is essential. When it comes to accuracy and productivity, deep learning-based systems outperform classical ones in image segmentation tasks. Image segmentation has made extensive use of convolutional neural networks, especially fully convolutional networks (FCNs). FCNs can directly produce pixel-wise segmentation maps by learning end-to-end mappings from input images to corresponding segmentation masks. For medical image segmentation, the widely used U-Net architecture combines a decoder-encoder framework with bypass links to gather both regionally and globally relevant data.

Deep learning-based segmentation models are often trained using annotated datasets that have been explicitly labelled by professionals. As part of the training phase, a damage function, such as dice damage or cross-entropy damage, is optimised to reduce the difference between the estimated segmentation maps and the actual labels. The robustness and generalisation of the segmentation models are frequently improved by using data enhancement methods including shifting, scaling, and mirroring [6]. Depending on the network design and level of supervision, multiple deep learning-based segmentation algorithms can be identified. For instance, fully supervised methods utilize pixel-level annotations for training, while weakly supervised and semi-supervised approaches use partial or incomplete annotations. Additionally, there are instance segmentation methods that aim to detect and segment individual instances of objects within the image.

The overlap and border agreement between the anticipated segmentation and the actual segmentation are measured using quantitative measures like the Dice coefficient, Jaccard index, and Hausdorff distance when evaluating image segmentation methods. Visual assessment and clinical validation are also essential for evaluating the accuracy and clinical relevance of the segmentation results [7].

Preprocessing techniques for medical images using deep learning have the ability to improve the effectiveness, precision, and reliability of numerous clinical imaging applications. They contribute to improving the accuracy of diagnosis, treatment planning, and monitoring of diseases and conditions. However, it is critical to address the constraints and difficulties of deep learning-based preprocessing techniques, such as the requirement for sizable annotated datasets, the interpretability of the models, and the possibility of excessive fitting. Future research directions should focus on

addressing these challenges and further refining and validating deep learning-based preprocessing techniques for various medical imaging modalities.

III. DEEP LEARNING-BASED PREPROCESSING TECHNIQUES

A. Denoising

1) *Convolutional Neural Networks (CNNs) for Denoising:* CNNs are frequently used to reduce noise in medical imaging. These networks are well-suited for capturing spatial information and learning complex relationships within the image data. In many CNN-based denoising models, the encoder collects hierarchical characteristics from the noisy source photograph, and the decoder reconstructs the denoised photograph using the learnt characteristics. During the reconstruction process, bypass links, which link corresponding layers in the encoder and decoder, aid in maintaining low-level features. These models are trained by minimising the difference between the denoised output and the clean target picture, using various loss coefficients like mean squared error (MSE) or perceptual loss [8].

2) *Generative Adversarial Networks (GANs) for Denoising:* By learning an inference network to create denoised images and a discriminant network to differentiate between the generated denoised photographs and the clean target photographs, Generative Adversarial Networks (GANs) have shown significant promise in denoising medical photographs. To produce denoised images with improved visibility and detail, GAN-based denoising models make use of the competitive learning process. While the discriminant network aids in directing the generator towards producing good-quality denoised photographs, the inference network seeks to generate genuine and visually comparable denoised photographs to the clean photos.

B. Image Enhancement

1) *Deep Learning-Based Contrast Enhancement:* The contrast of clinical photographs has been successfully increased using deep learning algorithms, which improves the visual appeal and interpretability of the photographs. The goal of deep learning-based contrast enhancement techniques is to figure out how to map low-contrast photographs to their respective improved ones. These methods leverage CNN architectures to extract image features and generate enhanced images that preserve the overall structure while highlighting important details. To train the networks, either paired low-contrast and high-contrast photographs or unsupervised methods like generative models are used. The training process involves minimizing specific loss functions to optimize the model parameters and achieve effective contrast enhancement [9].

2) *Super-Resolution Techniques:* Super-resolution techniques using deep learning have been employed to enhance the resolution and level of detail in low-resolution medical images. By studying underlying visual patterns and structures, these techniques seek to rebuild high-quality pictures from their low-quality equivalents. Deep learning-based super-resolution methods typically utilize CNN architectures, such as SRGAN, to generate high-resolution

images. After being trained on couples of poor quality and good quality photos, the networks may link the input and target images. Through the training process, the networks can generate high-resolution images with improved visual quality and enhanced details, which are beneficial for accurate diagnosis and analysis.

C. Image Registration

1) *Deep Learning-Based Registration Methods:* Significant improvements in aligning and matching medical pictures have been made using deep learning-based registration techniques. These methods utilize neural networks to predict transformation parameters or deformation fields that align the images. A couple of registered photographs with specified ground truth transformation information are used to train the networks. The network architectures typically involve convolutional layers for feature extraction and regression layers for predicting the transformation parameters. By learning the spatial transformations directly from the data, deep learning-based registration methods can handle complex deformations and achieve accurate alignment of medical images [10].

2) *Spatial Transformations and Neural Networks:* Spatial transformations combined with neural networks have been employed for image registration tasks. These methods utilize neural networks to predict spatial transformations or deformation fields that align the images. The network architectures typically consist of convolutional layers for feature extraction and regression layers for predicting the transformation parameters. The predicted transformations are then applied to the input images to achieve registration. These methods are advantageous as they can learn non-linear transformations, which are essential for handling complex deformations in medical images. By leveraging the representational learning capabilities of neural networks, these methods can effectively capture the spatial relationships and patterns within the images, leading to accurate image registration results [11].

D. Image Segmentation

1) *Convolutional Neural Networks for Segmentation:* CNNs have become an effective tool for segmenting clinical photographs. CNN-based segmentation techniques take advantage of deep networks' capacity to extract hierarchical characteristics from input images and produce pixel-by-pixel segmentation maps. These methods typically employ fully convolutional network architectures, where the fully connected layers of traditional CNNs are replaced with convolutional layers to enable dense pixel-wise predictions. The networks are trained on annotated datasets, where experts manually label the regions of interest in the images. To reduce the difference between the anticipated segmentation maps and the actual truth labels, the training procedure involves optimising specific loss parameters, such as cross-entropy damage or dice damage [12].

2) *U-Net and Variants for Segmentation:* In clinical picture segmentation work, U-Net and its derivatives have become increasingly popular. A bypass link between corresponding levels in the encoder and decoder is incorporated into the U-Net encoder-decoder architecture. A

precise and accurate segmentation is made possible by this design, which enables the network to collect both local and global contextual data. Numerous medical imaging applications, such as organ segmentation, tumour identification, and lesion segmentation, frequently use U-Net. Several extensions and variations of U-Net have been proposed to further improve its performance, such as using residual connections, introducing attention mechanisms, and incorporating multi-scale information [13].

IV. APPLICATIONS OF DEEP LEARNING-BASED PREPROCESSING

A. Computer-Aided Diagnosis (CAD)

Computer-Aided Diagnosis (CAD) refers to the use of computational algorithms and tools to aid radiology professionals and clinicians in the task of diagnosing diseases and conditions based on medical imaging data. Deep learning-based preprocessing techniques play a crucial role in CAD systems by improving the quality and reliability of the input images. The denoising, enhancement, and segmentation methods discussed earlier contribute to enhancing the visibility of lesions, abnormalities, and important anatomical structures, thereby aiding in the accurate interpretation and diagnosis of medical images. CAD systems can integrate these preprocessing techniques with advanced machine learning and artificial intelligence algorithms to automate or assist in the detection, characterization, and classification of various diseases [14].

B. Tumor Detection and Classification

Tumour identification and classification tasks have shown significant potential for deep learning-based preprocessing techniques. These techniques can enhance the visibility of tumors, remove noise and artifacts, and provide more precise segmentation of tumor regions. With improved image quality, deep learning models can effectively detect and classify tumors from medical images, including CT scans, MRIs, and histopathological images. CNN-based models, such as Faster R-CNN and YOLO, have been used for tumor detection, while CNN architectures like VGGNet, ResNet, and InceptionNet have been employed for tumor classification. These models can help radiologists and physicians in the proper diagnosis and treatment planning of various tumour types since they are trained on sizable annotated datasets to understand the distinguishing characteristics of various tumour types [15].

C. Image-Guided Interventions

Deep learning-based preprocessing techniques have applications in image-guided interventions, where medical images are used to guide and monitor surgical procedures and minimally invasive interventions. By improving image quality and enhancing relevant structures, these techniques provide more accurate and detailed guidance for surgeons and interventional radiologists. For example, denoising techniques can reduce the impact of noise during real-time imaging, enhancing the visualization of instruments, anatomical landmarks, and critical structures. Image registration techniques can align preoperative and intraoperative images, enabling accurate localization and navigation during procedures. Deep learning-based segmentation methods can delineate target structures and organs at risk, facilitating precise targeting and avoidance of critical areas during interventions.

D. Surgical Planning and Navigation

Deep learning-based preprocessing techniques have significant implications for surgical planning and navigation. By improving image quality and providing detailed anatomical information, these techniques enable surgeons to better understand patient anatomy and plan surgical approaches. Denoising and image enhancement techniques contribute to clearer and more informative images, allowing surgeons to visualize important structures and potential pathology. Image segmentation techniques aid in the identification and delineation of anatomical regions of interest and can guide surgical resection and target localization. These preprocessing techniques can be integrated into surgical navigation systems, providing real-time feedback and guidance during the procedure, thereby enhancing surgical accuracy and patient outcomes [16].

E. Radiomics and Quantitative Imaging

Radiomics and quantitative imaging involve the collection of numerical information from clinical photographs in order to characterize diseases, assess treatment responses, and predict patient outcomes. Deep learning-based preprocessing techniques can contribute to the accurate extraction and analysis of these features. By enhancing image quality, reducing noise, and improving segmentation, these techniques enable more robust and reliable quantification of imaging biomarkers. Deep learning algorithms can be trained using enhanced images, and they will subsequently extract characteristics and perform analysis. Deep learning-based feature extraction methods can capture subtle patterns and textures that are indicative of disease characteristics and treatment responses. Then, result forecasting, treatment scheduling, and risk classification can be done using these derived features [17].

V. EVALUATION AND PERFORMANCE METRICS

A. Quantitative Evaluation of Deep Learning-Based Preprocessing Techniques

The quantitative evaluation of deep learning-based preprocessing techniques is essential to assess their effectiveness and compare different approaches. Different criteria are frequently employed to assess how well these strategies operate. Peak signal-to-noise ratio (PSNR) and structural similarity index measures (SSIM) are widely used in denoising operations to assess the calibre and similarity of the denoised image to the clean ground truth image. Additionally, the pixel-level variations between the denoised and clean images can be measured using mean squared error (MSE) and root mean square error (RMSE).

In the context of image enhancement, metrics like contrast improvement, entropy, and subjective evaluations by radiologists can be utilized to assess the visual quality and improvement achieved by the preprocessing techniques. Objective metrics such as the mean enhancement index (MEI) and signal-to-noise ratio (SNR) can provide quantitative measures of the enhancement achieved.

The accuracy of the alignment of the registered pictures is frequently assessed for image registration using metrics like the Dice coefficient, Jaccard index, and Hausdorff distance. These metrics provide an accurate measurement of the overlap and separation between the registered image and the reference image [18].

Dice coefficient, Jaccard index, sensitivity, specificity, and accuracy are a few of the evaluation measures for image segmentation that gauge how closely and consistently the projected segmentation matches the actual annotations on the ground. The reliability and accuracy of the segmentation findings are revealed by these measures. It's crucial to remember that the selection of evaluation measures ought to match the preprocessing technique's specific purpose and goals. Additionally, it is advised to take into account a variety of evaluation measures to get a complete picture of the effectiveness of the strategy.

B. Comparative Studies and Benchmark Datasets

Comparative studies and benchmark datasets are crucial for assessing and comparing the performance of deep learning-based preprocessing techniques. Comparative studies involve evaluating multiple techniques on the same datasets and using the same evaluation metrics to facilitate fair comparisons. These studies can help identify the strengths and weaknesses of different approaches and provide insights into their relative performance.

Benchmark datasets are essential for assessing preprocessing methods. These datasets should be diverse, representative of the clinical scenarios, and contain ground truth annotations for accurate evaluation. They make it possible for academics to assess their methods using standardised datasets and analyse the results against those obtained using cutting-edge methodologies. Examples of well-known benchmark datasets in the field of diagnostic imaging include the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, the Lung Image Database Consortium (LIDC) dataset for lung cancer imaging, and the Medical Image Computing and Computer Assisted Intervention (MICCAI) challenge datasets [19].

The availability of benchmark datasets promotes reproducibility, makes it easier to compare various methods, and gives researchers a shared platform to assess and improve their methods. Nevertheless, it is critical to consider the prejudices and constraints associated with benchmark datasets, as they may not fully represent the variability and complexity encountered in real-world clinical scenarios.

C. Limitations and Challenges in Evaluation

There are several limitations and challenges to evaluating deep learning-based preprocessing techniques for medical images. One of the challenges is the lack of ground truth annotations for certain tasks, such as subjective image quality enhancement or the removal of artifacts. In such cases, subjective evaluations by expert radiologists or clinicians become crucial for assessing the effectiveness of the techniques. Another challenge is the potential overfitting of the models to specific datasets, which can restrict the scope of the outcomes. It is important to validate the performance of the techniques on independent datasets to ensure their robustness and applicability in different clinical settings.

Furthermore, the selection of appropriate evaluation metrics can be challenging, as different metrics may focus on different aspects of performance. It is important to choose metrics that align with the specific task and provide meaningful insights into the performance of the preprocessing techniques. Careful consideration should be given to metrics that capture relevant aspects such as image quality, accuracy, segmentation overlap, or clinical utility.

The interpretability of evaluation results is another challenge. While quantitative metrics provide objective measures of performance, they may not fully capture the clinical impact or the perceptual quality of the preprocessed images. Qualitative assessments and expert evaluations by radiologists or clinicians are often necessary to complement quantitative metrics and provide a comprehensive evaluation.

Moreover, the reproducibility of results is an important consideration in the evaluation of deep learning-based preprocessing techniques. The choice of hyperparameters, network architectures, and training strategies can significantly impact performance. Therefore, detailed documentation of experimental setups and open-sourcing of code and models are encouraged to facilitate reproducibility and promote transparency in the research community.

Finally, it is essential to address the limitations and challenges specific to the domain of medical imaging. Medical images often exhibit inherent variability due to differences in imaging protocols, acquisition devices, patient populations, and pathological conditions. This variability poses challenges in developing preprocessing techniques that generalize well across diverse datasets and clinical scenarios. Future research should focus on addressing these challenges through the development of robust and adaptable preprocessing techniques that can effectively handle the variability encountered in medical images.

VI. FUTURE DIRECTIONS AND EMERGING TRENDS

A. Interpretability and Explainability in Deep Learning Preprocessing

Interpretability and explainability of deep learning models and preprocessing techniques are crucial for their adoption and trustworthiness in clinical settings. As deep learning models become increasingly complex, understanding the reasoning behind their decisions becomes more challenging. Further research should concentrate on creating methods that give comprehensible justifications for the preprocessing steps and their impact on the final results. This can involve the use of attention mechanisms, visualization techniques, or model-agnostic approaches to provide insights into the important features and regions that contribute to the preprocessing outcomes. Explainable preprocessing methods will not only enhance the transparency of the techniques but also enable radiologists and clinicians to have better insights into the decision-making process and facilitate more confident clinical decision-making [20].

B. Multimodal Image Preprocessing

With the advent of multimodal medical imaging, such as the fusion of CT and MRI, there is a need for preprocessing techniques that can effectively handle and integrate information from multiple imaging modalities. Future research should focus on developing deep learning-based preprocessing methods that can extract complementary and synergistic information from multimodal images. These methods can boost the reliability and accuracy of tasks involving image fusion, registration, and segmentation. Additionally, the integration of multimodal preprocessing with deep learning models can enhance the performance of subsequent analysis and decision-making processes.

C. Transfer Learning and Domain Adaptation

Transfer learning and domain adaptation techniques have shown promise for improving the generalizability and applicability of deep learning models. These techniques enable the transfer of knowledge learned from one domain or dataset to another, even when the target domain has limited labelled data. In the context of preprocessing, transfer learning can be leveraged to initialize deep learning models with pre-trained weights from large-scale datasets and then fine-tune them on smaller, domain-specific medical imaging datasets. This approach can help overcome the challenge of limited annotated medical image datasets and improve the performance and efficiency of preprocessing techniques. Future research should explore transfer learning and domain adaptation techniques specifically tailored for medical image preprocessing to enable the effective utilization of existing knowledge and promote the development of robust and adaptive preprocessing methods [21].

D. Real-Time Preprocessing for Clinical Applications

Real-time preprocessing is of paramount importance for clinical applications where immediate and on-the-fly processing of medical images is required. In scenarios such as image-guided interventions or real-time monitoring, preprocessing techniques need to operate in real-time to provide timely and accurate results. Future research should focus on developing efficient and optimized preprocessing algorithms that can handle the computational demands of real-time processing. Techniques such as model compression, hardware acceleration, and parallel computing can be explored to enable fast and efficient preprocessing without compromising the quality and accuracy of the results. Real-time preprocessing will significantly enhance the clinical utility of deep learning-based techniques by providing immediate feedback and facilitating timely decision-making during interventions, surgeries, and other time-critical procedures.

VII. CONCLUSION

In the fields of computer science and medical imaging, the preprocessing of medical images using deep learning techniques has become a powerful and promising topic of research. This study has provided a comprehensive overview of various aspects related to the preprocessing of medical images using deep learning. The background section highlighted the significance of medical image preprocessing in enhancing the quality, accuracy, and clinical utility of medical imaging. It discussed the challenges associated with medical image preprocessing, such as noise reduction, image enhancement, image registration, and image segmentation. The motivation section emphasized the potential benefits of deep learning-based preprocessing techniques, including improved image quality, increased diagnostic accuracy, and enhanced clinical decision-making. The objectives of this review were to explore and discuss the advancements in deep learning-based preprocessing techniques and their application in medical imaging. The study detailed various preprocessing tasks, including denoising, image enhancement, image registration, and image segmentation, and provided an in-depth analysis of the role of deep learning in each task. The review study also highlighted the quantitative evaluation of deep learning-based preprocessing techniques and discussed the importance of comparative studies and benchmark datasets in assessing their performance. It addressed the limitations and challenges of evaluating these techniques,

such as the availability of ground truth annotations, the interpretability of results, and reproducibility. Furthermore, the study discussed future directions and emerging trends in the field of deep learning-based medical image preprocessing. It emphasized the importance of interpretability and explainability in deep learning models, the need for multimodal image preprocessing techniques, the potential of transfer learning and domain adaptation, and the significance of real-time preprocessing for clinical applications.

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